

Predicting Colour Yield from Process Control Limits

A physics-based capability study of anodizing and PVD thin-film finishing

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Abstract

Manufactured products that must look identical — a laptop lid, a watch case, a phone frame — get their colour from a surface finishing process, and that process drifts. This study builds a physics model of two industrial finishing routes, anodizing-and-dyeing and physical vapour deposition (PVD), and pushes realistic process variation through them to predict what fraction of parts land inside a colour specification. The two routes are found to fail in entirely different places. A black anodized finish is almost immune to colour variation ($C_{pk} = 16.7$) and is limited instead by film thickness ($C_{pk} = 1.80$). A PVD interference stack has no thickness problem at all, yet yields only 56.4% at realistic tolerances, with a single variable — nitrogen stoichiometry — responsible for 69.8% of the colour variance. Solving for the control limits that would achieve $C_{pk} \geq 1.33$ requires tightening nitrogen control sixfold. A third result is that the colour a designer selects is itself a capability decision: holding the anodizing process completely fixed and changing only the dye moves yield from 100% to 77.7%.

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1 The problem, in plain language

1.1 Why this matters

Pick up any anodized aluminium laptop or any coated stainless steel watch. It has a colour. Now consider that the factory made a million of them, and every single one has to be the *same* colour — close enough that a customer holding two of them side by side cannot tell.

That is harder than it sounds, because colour is not painted on. It is the by-product of a chemical or physical process running in a tank or a vacuum chamber, and every one of those processes drifts. The acid bath warms up over a shift. The dye gets consumed. A gas flow controller reads half a percent high. None of these drifts is a fault; they are the normal breathing of a production line. The question is whether the colour they produce still falls inside the tolerance.

The question this study answers. Given how tightly a real production line holds its process settings, what fraction of parts come out the right colour — and if the answer is not good enough, which single control should be tightened first?

This matters commercially because cosmetic rejects are expensive and late. A part that fails a dimensional check fails at the machine. A part that fails a *cosmetic* check fails at the very end, after all the value has been added, and it usually cannot be reworked.

1.2 What gets built

The study builds a computer model that runs the whole causal chain:

process settings drift → the film that grows changes → the light it reflects changes
→ the colour a human sees changes → some parts fall outside spec

Because every link is physics rather than a curve fit, the model can be asked questions no lookup table could answer — for example, what happens at a temperature nobody has tried yet.

1.3 A glossary, for readers new to the field

Anodizing

An electrochemical process. An aluminium part is made the positive electrode in an acid bath, and a hard, porous oxide skin grows out of the metal's own surface. The pores can then be filled with dye, which is how anodized parts get colour. Used on most aluminium consumer electronics.

PVD (physical vapour deposition)

A vacuum process. A solid metal target is vaporised, and the vapour condenses on the part as an extremely thin film — typically a few hundred nanometres, about a thousandth of a human hair. Reacting the vapour with nitrogen gas makes titanium nitride, a hard ceramic that happens to be gold-coloured. Used for watch cases, taps, and tooling.

Reflectance spectrum, $R(\lambda)$

The fraction of light a surface reflects, measured separately at each wavelength λ . This is the raw physical description of colour: an object looks red because its spectrum is high in the red and low elsewhere.

CIELAB

A colour coordinate system with three numbers: L^* for lightness (0 black, 100 white), a^* for green-to-red, b^* for blue-to-yellow. It is designed so that equal distances correspond roughly to equally visible differences.

ΔE_{00} (CIEDE2000)

The standard measure of how different two colours look. Roughly, $\Delta E_{00} \approx 1$ is the smallest difference a trained observer can reliably see. Cosmetic specifications for consumer products are usually set at or below this value.

C_{pk} (process capability index)

A single number saying how comfortably a process fits inside its tolerance. $C_{pk} = 1.0$ means the tolerance band is exactly as wide as the process spread; $C_{pk} = 1.33$ is the usual consumer electronics gate and corresponds to roughly 32 defective parts per million. Below 1.0 the process is not fit for the specification.

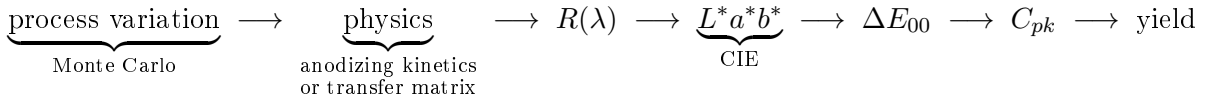
DOE (design of experiments)

A structured plan for varying several settings at once, so that their individual and combined effects can be separated from a small number of runs rather than a huge number.

2 Method

2.1 The shared pipeline

Both finishing routes are modelled independently up to the point where they produce a reflectance spectrum, after which they share one common path to colour and capability:



Sharing the colour engine is what makes the two routes directly comparable: any difference in the final capability number is attributable to the physics of the finish, not to differences in how colour was computed.

2.2 From a spectrum to a colour

Human colour vision has three receptor types. The CIE formalises this with three *colour matching functions* $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$, which describe the sensitivity of a standard observer. Multiplying the reflectance by the illuminating light $S(\lambda)$ and by each of these, then integrating, yields the tristimulus values:

$$X = k \int R(\lambda) S(\lambda) \bar{x}(\lambda) d\lambda, \quad k = \frac{100}{\int S(\lambda) \bar{y}(\lambda) d\lambda} \quad (1)$$

with Y and Z defined analogously. The illuminant used throughout is D65, standard average daylight. XYZ is then converted to CIELAB through a cube-root compression that mimics the eye's non-linear lightness response, and two colours are compared using the CIEDE2000 formula,

$$\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \frac{\Delta C'}{k_C S_C} \frac{\Delta H'}{k_H S_H}} \quad (2)$$

whose weighting terms S_L, S_C, S_H and rotation term R_T correct for the fact that the eye's sensitivity depends on where in colour space the comparison is being made — it is far more sensitive to a shift in a pale grey than to the same numerical shift in a saturated blue.

2.3 Route 1: anodizing

Anodic oxide grows by Faraday's law from the anodizing current, while the acid simultaneously dissolves it back. The net growth is the difference:

$$\frac{dx}{dt} = \underbrace{\frac{M_{\text{ox}} j \eta}{z F \rho_{\text{ox}}}}_{\text{electrochemical growth}} - \underbrace{k_0 \exp\left(-\frac{E_a}{RT}\right) [\text{H}_2\text{SO}_4]^m}_{\text{chemical back-dissolution}} \quad (3)$$

with $M_{\text{ox}} = 101.96$ g/mol for Al_2O_3 , $z = 6$ (forming one Al_2O_3 unit transfers six electrons), and $\rho_{\text{ox}} \approx 3.1$ g/cm³ — below dense sapphire, because the anodic film is porous and partly hydrated.

Why bath temperature dominates. The second term in Eq. (3) is Arrhenius, so it roughly doubles for every +10°C. A bath that drifts warm therefore does two bad things at once: it thins the film, and it widens the pores. The part comes out lighter and duller. No other single variable has this double effect, which is why temperature control is the first thing any anodizing line invests in.

The dye lives in the pores, so dye loading scales with pore volume, that is, with porosity multiplied by thickness. Uptake saturates exponentially with immersion time and peaks near pH 5.5, where acid dyes are most readily adsorbed by the amphoteric oxide.

A dyed anodic film is a *bulk absorber*, not an interference filter. Light crosses the film, reflects off the aluminium underneath, and crosses back, so the Beer–Lambert law applies over a double pass:

$$R(\lambda) = R_{\text{surf}} + (1 - R_{\text{surf}})^2 R_{\text{Al}}(\lambda) e^{-2\varepsilon(\lambda) c x} \quad (4)$$

Because there is no interference term, dyed anodizing barely changes colour with viewing angle — the exact opposite of the PVD stack considered next.

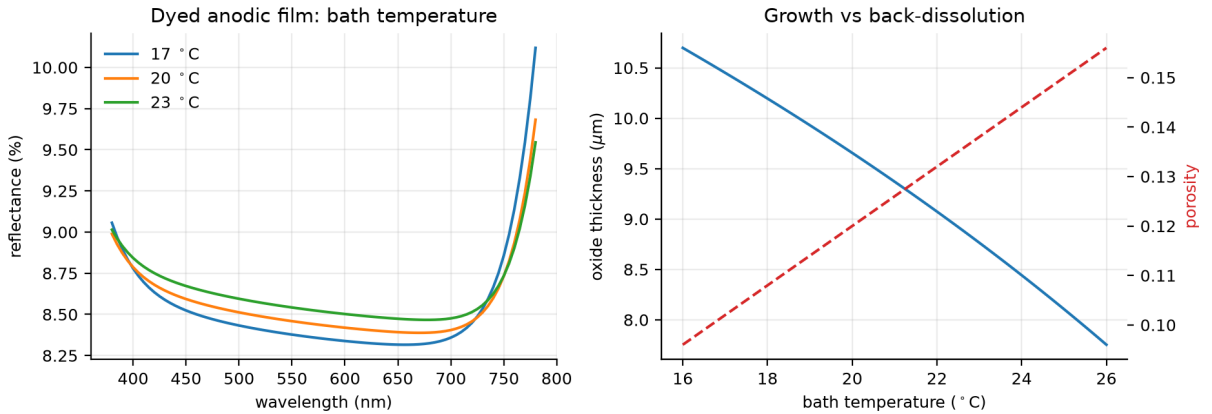


Figure 1: Left: the reflectance of a black-dyed film across a $\pm 3^\circ\text{C}$ bath excursion. Right: the competition in Eq. (3) — as the bath warms, back-dissolution accelerates and thickness falls (blue), while porosity rises (red, dashed).

2.4 Route 2: PVD thin films

A PVD coating is a stack of layers each a few hundred nanometres thick, which is comparable to the wavelength of visible light. Light reflected from the top of a layer therefore interferes with light reflected from its bottom, and the colour depends on the whole stack rather than on any one surface.

The standard treatment assigns each layer a characteristic matrix,

$$M_j = \begin{bmatrix} \cos \delta_j & \frac{i \sin \delta_j}{\eta_j} \\ i \eta_j \sin \delta_j & \cos \delta_j \end{bmatrix}, \quad \delta_j = \frac{2\pi t_j N_j \cos \theta_j}{\lambda} \quad (5)$$

where $N_j = n_j - ik_j$ is the complex refractive index and η_j the tilted optical admittance. Multiplying the matrices maps the substrate admittance to the front surface, giving

$$\begin{bmatrix} B \\ C \end{bmatrix} = \left(\prod_j M_j \right) \begin{bmatrix} 1 \\ \eta_{\text{sub}} \end{bmatrix}, \quad Y = \frac{C}{B}, \quad R = \left| \frac{\eta_0 - Y}{\eta_0 + Y} \right|^2 \quad (6)$$

This is the same calculation performed by the commercial thin-film design packages Essential Macleod and TFCALC.

A sign convention that silently destroys the answer. Tabulated optical constants are published as $N = n + ik$. The matrix formulation above expects $N = n - ik$. Feeding the first into the second turns every absorbing layer into a *gain* medium: the code runs, the plots look plausible, and reflectance quietly exceeds 100%. This was caught here by a test asserting $0 \leq R \leq 1$ for every stack at every angle, and that test is retained precisely because the failure is invisible without it.

The colour of a bulk nitride is controlled by its *composition*, not its thickness. Titanium nitride absorbs strongly enough that past roughly 150 nm the film is optically opaque and the substrate beneath it becomes irrelevant. The variable that actually matters is the stoichiometry x in TiN_x , set on the tool by nitrogen partial pressure. As x falls below one the film is nitrogen-starved and metal-rich: more free carriers raise the plasma frequency, the reflectance edge moves blue, and the gold drifts toward metallic silver.

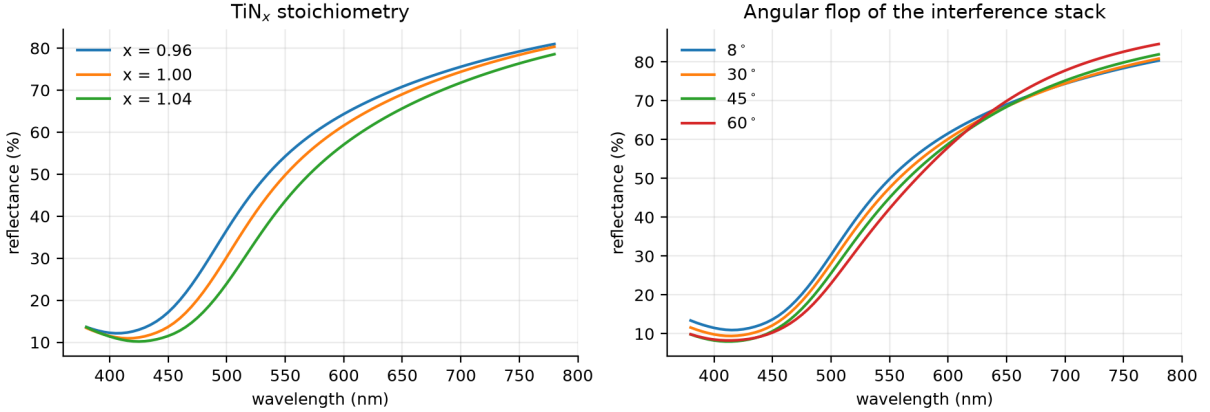


Figure 2: Left: reflectance of the PVD stack across a $\pm 4\%$ swing in nitrogen stoichiometry. Right: angular colour shift (“flop”) of the same interference stack — the reason a part can pass inspection face-on and fail at a chamfer.

2.5 Propagating variation

Each process input is assigned a control tolerance, interpreted as a $\pm 3\sigma$ band, and sampled from a normal distribution. Every sample is pushed through the forward model. Capability follows as

$$C_{pk} = \frac{\min(\text{USL} - \mu, \mu - \text{LSL})}{3\sigma}, \quad C_{pk}^{\text{upper}} = \frac{\text{USL} - \mu}{3\sigma} \quad (7)$$

the second form applying to ΔE_{00} , for which there is no such thing as too little colour error. Inputs are ranked by standardised regression coefficients, which give the response change per standard deviation of input and therefore already fold in both the physical sensitivity and the current control spread. That is the correct ranking for deciding which tolerance to buy down.

2.6 Experimental design

A Box–Behnken design is used throughout. Unlike a central composite design it places no run outside the stated factor range, which matters for a real bath: axial points would demand acid concentrations the line cannot hold. A full quadratic response surface is fitted on coded levels, so that coefficient magnitudes are directly comparable.

3 Verification

Before any result is reported, the implementation is checked against published values. The suite contains 62 tests; the load-bearing ones are:

Check	Result
All 13 CIEDE2000 reference pairs of Sharma, Wu & Dalal (2005)	Reproduced to within 10^{-4} . These pairs exist specifically to exercise the hue-wraparound and rotation terms that naive implementations get wrong.
Perfect reflecting diffuser under D65	Returns exactly $L^* = 100$, $a^* = b^* = 0$.
D65 white point	(95.047, 100, 108.883), matching the CIE tabulation.
$0 \leq R \leq 1$ for every stack at every angle	Passes; this is the check that caught the sign-convention error.
Nominal Type II recipe (1.5 A/dm ² , 30 min, 20°C, 180 g/L)	Returns 9.66 μm against the $\sim 10 \mu\text{m}$ industry specification. This is the calibration anchor for the entire anodizing model.
Back-dissolution temperature dependence	Verified to fall between $1.8\times$ and $3.0\times$ per $+10^\circ\text{C}$.
Response-surface fitter on a known quadratic	Recovers every coefficient exactly, $R^2 = 1$.

Table 1: Verification summary. All checks are automated and run in continuous integration.

4 Results

4.1 Anodizing: colour is easy, thickness is not

At the nominal window with a black dye, the process is comfortably capable on colour and marginal on thickness.

Response	Mean	σ	Spec	C_{pk}
ΔE_{00} vs colour master	0.032	0.019	≤ 1.0	17.04
Oxide thickness	9.653 μm	0.307 μm	8–12 μm	1.80
Combined yield				100.00 %

Table 2: Anodizing capability, $n = 6000$ Monte Carlo samples.

The Box–Behnken response surface fits thickness essentially perfectly ($R^2 = 0.99999$) and colour well ($R^2 = 0.932$). The thickness model is dominated by linear terms in current density, bath temperature and acid concentration, exactly as Eq. (3) demands.

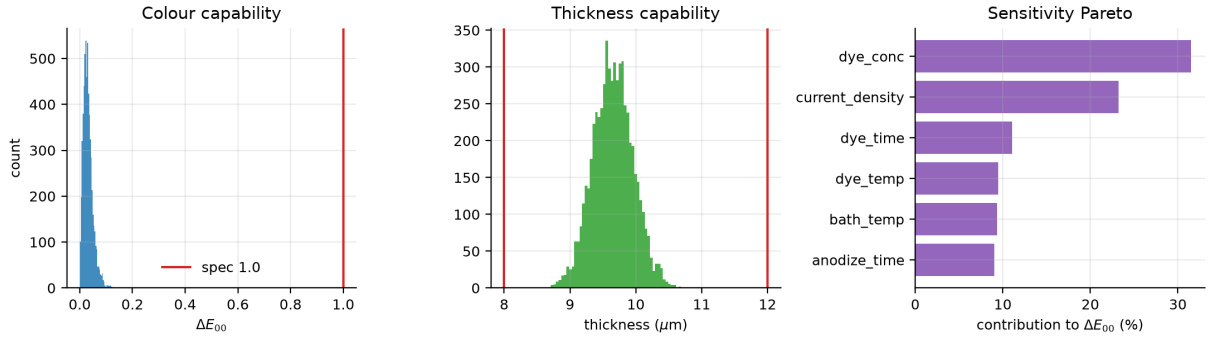


Figure 3: Anodizing capability. Colour (left) sits far inside its limit; thickness (centre) consumes most of its window; the Pareto (right) identifies dye concentration and current density as the leading contributors to colour variation.

4.2 The colour itself is a capability decision

The most consequential result in this study comes from changing nothing about the process. Holding the window, the tolerances and the sample size fixed, and varying only which dye is in the tank:

Dye	Mean ΔE_{00}	σ	C_{pk}	Yield
Black	0.032	0.019	16.68	100.00 %
Graphite grey	0.203	0.158	1.68	99.96 %
Gold / bronze	0.549	0.415	0.36	85.20 %
Red	0.566	0.427	0.34	84.24 %
Blue	0.668	0.512	0.22	77.72 %

Table 3: Identical process, identical tolerances, different dye.

A saturated black sits deep in Beer–Lambert saturation: the exponential has already bottomed out, so bath noise moves the reflectance hardly at all. Pale and chromatic finishes sit on the steep part of the same exponential, where the same noise produces a visible shift.

Implication. Yield swings by 22 percentage points with no process change whatsoever. The decision that caused it was made in a design review, by someone choosing a colour, long before any process engineer saw the part. This argues for capability analysis being run *at colour selection*, not after the line is commissioned.

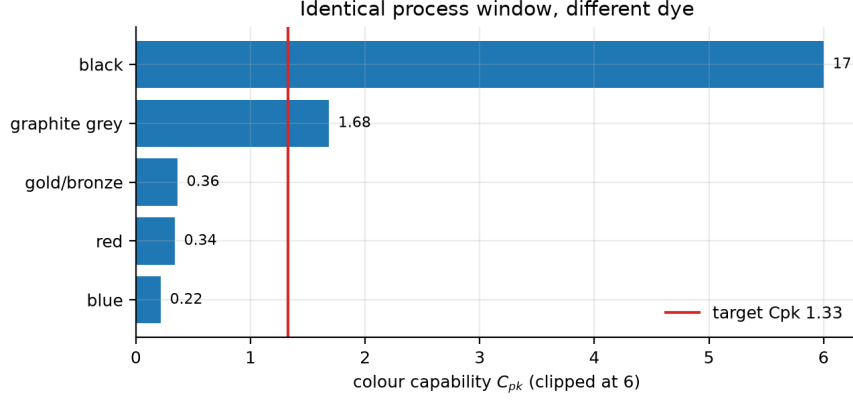


Figure 4: Colour capability by dye choice under an identical process window. Only black and graphite clear the $C_{pk} = 1.33$ gate.

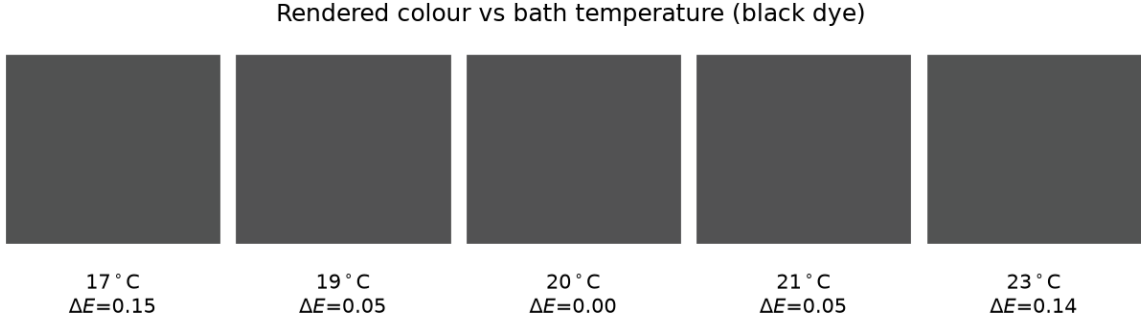


Figure 5: Rendered appearance across a $\pm 3^\circ\text{C}$ bath excursion, with the corresponding ΔE_{00} . Swatches are computed from the simulated spectra through the same CIE pipeline, then converted to sRGB for display.

4.3 PVD: one variable dominates, and it is not thickness

The PVD interference stack fails at tolerances that would be entirely normal on a production tool.

Quantity	Value	Variance share
Mean ΔE_{00}	1.042	Stoichiometry 69.8 %
$\sigma_{\Delta E_{00}}$	0.773	Overcoat thickness 27.8 %
Colour C_{pk}	-0.018	TiN thickness 2.5 %
Yield against $\Delta E_{00} \leq 1.0$	56.4 %	

Table 4: PVD capability for the stack air / SiO₂ 90 nm / TiN 300 nm / Ti 80 nm / SS304, $n = 6000$.

A negative C_{pk} means the process mean already sits outside the specification limit — the average part is a reject. The response surface for ΔE_{00} fits well ($R^2 = 0.996$, adjusted 0.989) and is dominated by *quadratic* terms in stoichiometry, meaning the colour penalty grows faster than linearly as nitrogen control drifts in either direction.

Greedily tightening the dominant tolerance until $C_{pk} \geq 1.33$ converges in 18 rounds:

Control	Current	Required	Change
TiN stoichiometry (nitrogen flow)	± 0.040	± 0.0067	$6.0\times$ tighter
SiO ₂ overcoat thickness	± 8.0 nm	± 3.0 nm	$2.7\times$ tighter
TiN thickness	± 20 nm	± 20 nm	unchanged

Table 5: Control limits required to reach $C_{pk} = 1.33$.

The counter-intuitive part. TiN thickness is the variable an engineer would instinctively control, and it is almost irrelevant — it accounts for 2.5 % of the colour variance and its tolerance never needs tightening. The film is optically opaque past roughly 150 nm, so adding more of it changes nothing an observer can see. Composition is the real lever, and composition is controlled by a gas flow, not by a deposition timer.

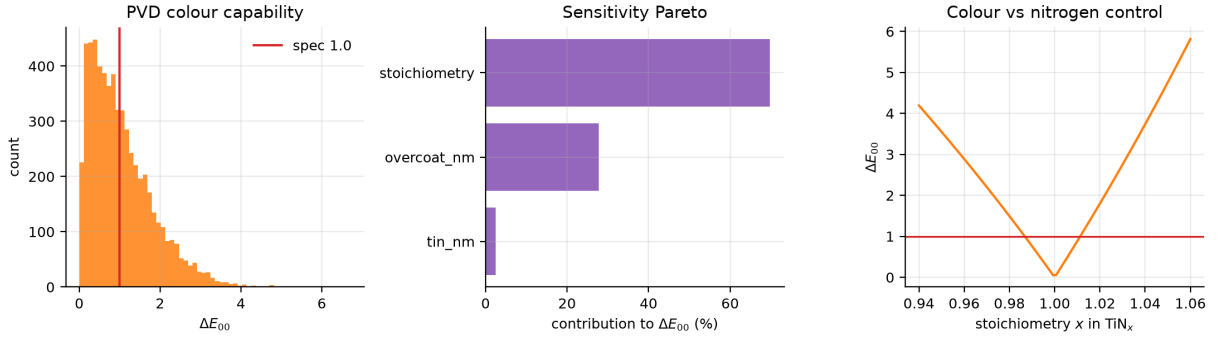


Figure 6: PVD capability. A large fraction of the simulated distribution (left) lies beyond the specification. Stoichiometry dominates the Pareto (centre). The colour penalty versus nitrogen control (right) is steeply quadratic, so symmetric drift in either direction is punished.

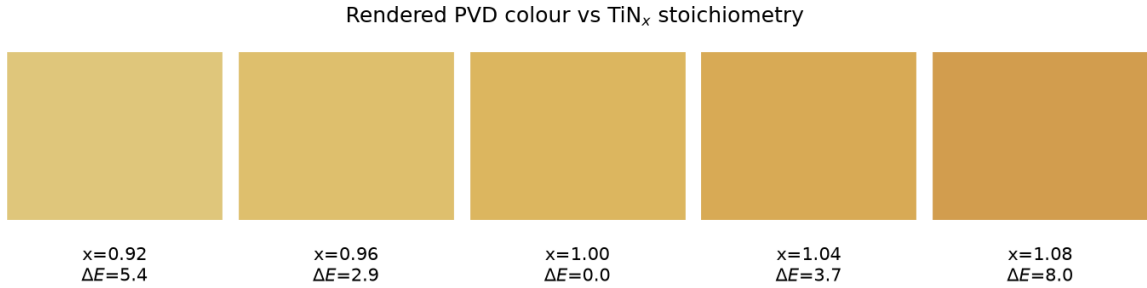


Figure 7: Rendered PVD appearance across the stoichiometry range. A $\pm 8\%$ composition swing takes the finish from pale champagne to warm bronze.

4.4 The two routes fail differently

	Anodize + black dye	PVD interference stack
Binding constraint	Film thickness	Composition
Colour C_{pk}	16.68	-0.018
Limiting C_{pk}	1.80 (thickness)	-0.018 (colour)
Dominant variable	Current density, dye conc.	Nitrogen stoichiometry
Angular colour shift	Negligible (bulk absorber)	Significant (interference)
What to instrument	Bath temperature, coating thickness gauge	Mass-flow control, optical monitor

Table 6: Comparative failure modes.

A control plan written for one route is close to useless for the other. The anodizing line needs thickness metrology and temperature control; the PVD line needs gas-flow precision and in-chamber optical monitoring. Neither conclusion is reachable without modelling the physics — both routes produce “a coloured part”, and their process sheets look superficially similar.

5 Limitations

These results are simulation. Stating the boundaries precisely is part of the result.

1. **Optical constants are representative literature fits**, not measurements of a specific supplier’s coating. They reproduce the correct colour family and the correct dispersion shape, which is what a *sensitivity* study requires: the conclusions concern how strongly colour responds to process variation, and that is governed by stack geometry and the shape of $n(\lambda)$ rather than by the third decimal place of any oscillator parameter. Absolute colour prediction against a physical master would require measured n, k data, which the code accepts as a drop-in replacement.
2. **The dye-uptake and porosity relations are phenomenological**. They are fitted to published Type II behaviour, not derived from first principles. Trends are trustworthy; absolute dye loadings are not.
3. **Capability here is short-term**. Samples are drawn at fixed nominal settings, so what is computed is within-batch capability. Real long-term P_{pk} on a line will be worse, because it additionally carries drift between bath changes, rack position, and operator.
4. **No experimental validation**. No coupons were anodized and no spectrophotometer readings were taken. This is the single largest gap, and the natural next step. The code is deliberately structured so that a measured dataset drops directly into the existing comparison path.

6 Conclusions

1. A single physics-to-colour-to-capability pipeline can serve two physically unrelated finishing routes, making them directly comparable on the only metric that matters commercially — yield against a colour specification.
2. The PVD interference stack studied here is not capable at realistic tolerances (56.4% yield, $C_{pk} = -0.018$). Reaching $C_{pk} = 1.33$ requires sixfold tighter nitrogen control, and no improvement whatsoever in the thickness control an engineer would instinctively tighten first.
3. Anodized black is colour-robust and thickness-limited; the constraint and therefore the entire control strategy is the opposite of the PVD case.

4. Cosmetic capability should be assessed at the point of *colour selection*. Changing only the dye, with the process held completely fixed, moved yield from 100 % to 77.7 %. That is a larger effect than most process improvements could recover, and it is free to avoid if it is considered early enough.

Reproducibility

Every number and figure in this report is regenerated by a single command:

```
python scripts/run_study.py
```

The test suite (`python -m pytest`, 62 tests) runs in continuous integration on Python 3.10–3.12, and the study itself is re-executed on every push, so the document and the code cannot drift apart. The design tables are also exported as JMP-ready CSV together with a JSL script that reconstructs the response-surface analysis, prediction profiler and contour profiler directly.

References

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